Task 1: Bank ATM System

Scenario

An ATM allows customers to perform basic operations such as checking their balance, depositing money, and withdrawing money.

Requirements

Create a Java program using **separate methods** for each operation.

Implement the following methods:

- `checkBalance()`
- `deposit(double amount)`
- `withdraw(double amount)`
- `displayMenu()`

Conditions

- Initial balance = ₹20,000.
- Deposit should increase the balance.
- Withdrawal should happen only if sufficient balance is available.
- Display the updated balance after every transaction.
- Call the methods from `main()`.

Task 2: Student Result Management System

Scenario

A school wants to automate student result processing.

Requirements

Create methods to perform the following tasks:

- `acceptMarks()`
- `calculateTotal()`
- `calculateAverage()`
- `findGrade()`
- `displayResult()`

Conditions

- Read marks of 5 subjects.
- Calculate total and average.
- Assign grades based on average:
  - 90–100 → A+
  - 80–89 → A
  - 70–79 → B
  - 60–69 → C
  - Below 60 → Fail
- Display all details using a separate method.

Task 3: Online Food Ordering System

Scenario

A food delivery application calculates the customer's bill after ordering food.

Requirements

Create methods for:

- `displayMenu()`
- `calculateBill(int quantity, double price)`
- `calculateDiscount(double bill)`
- `printBill()`

Conditions

- Display at least 5 food items with prices.
- Allow the user to choose one item and enter the quantity.
- Calculate the total bill.
- Apply a **10% discount** if the bill exceeds ₹1000.
- Print the final bill using a separate method.
- Call all methods from `main()`.

Task 1: Food Delivery App (Swiggy/Zomato)

Scenario

A food delivery application allows customers to order food from different restaurants.

Requirements

Create a class named `FoodOrder`.

Data Members

- Order ID
- Customer Name
- Restaurant Name
- Food Item
- Quantity
- Price per Item

Constructor

Use a parameterized constructor to initialize all order details.

Methods

Create methods to:

- Calculate the total bill.
- Apply a discount if the bill exceeds ₹1000.
- Calculate the delivery charge.
- Display the final bill.

Conditions

- If the bill is greater than ₹1000, apply a **15% discount**.
- If the bill is less than ₹500, add a **₹50 delivery charge**.
- Otherwise, delivery is free.

In `main()`

- Create details for **3 customers** using the constructor.
- Display each customer's final bill.

Task 2: OTT Subscription System (Netflix/Prime/Hotstar)

Scenario

An OTT platform allows users to subscribe to different plans.

Requirements

Create a class named `Subscription`.

Data Members

- User ID
- User Name
- Platform Name
- Subscription Plan
- Monthly Price
- Number of Months

Constructor

Initialize all values using a parameterized constructor.

Methods

Create methods to:

- Calculate the subscription amount.
- Apply a promotional discount.
- Calculate GST.
- Display the final payable amount.

Conditions

- If the subscription is for **12 months**, give a **20% discount**.
- Add **18% GST** after applying the discount.
- Display all subscription details and the final amount.

In `main()`

- Create **3 user subscriptions**.
- Display the payment details for each user.
